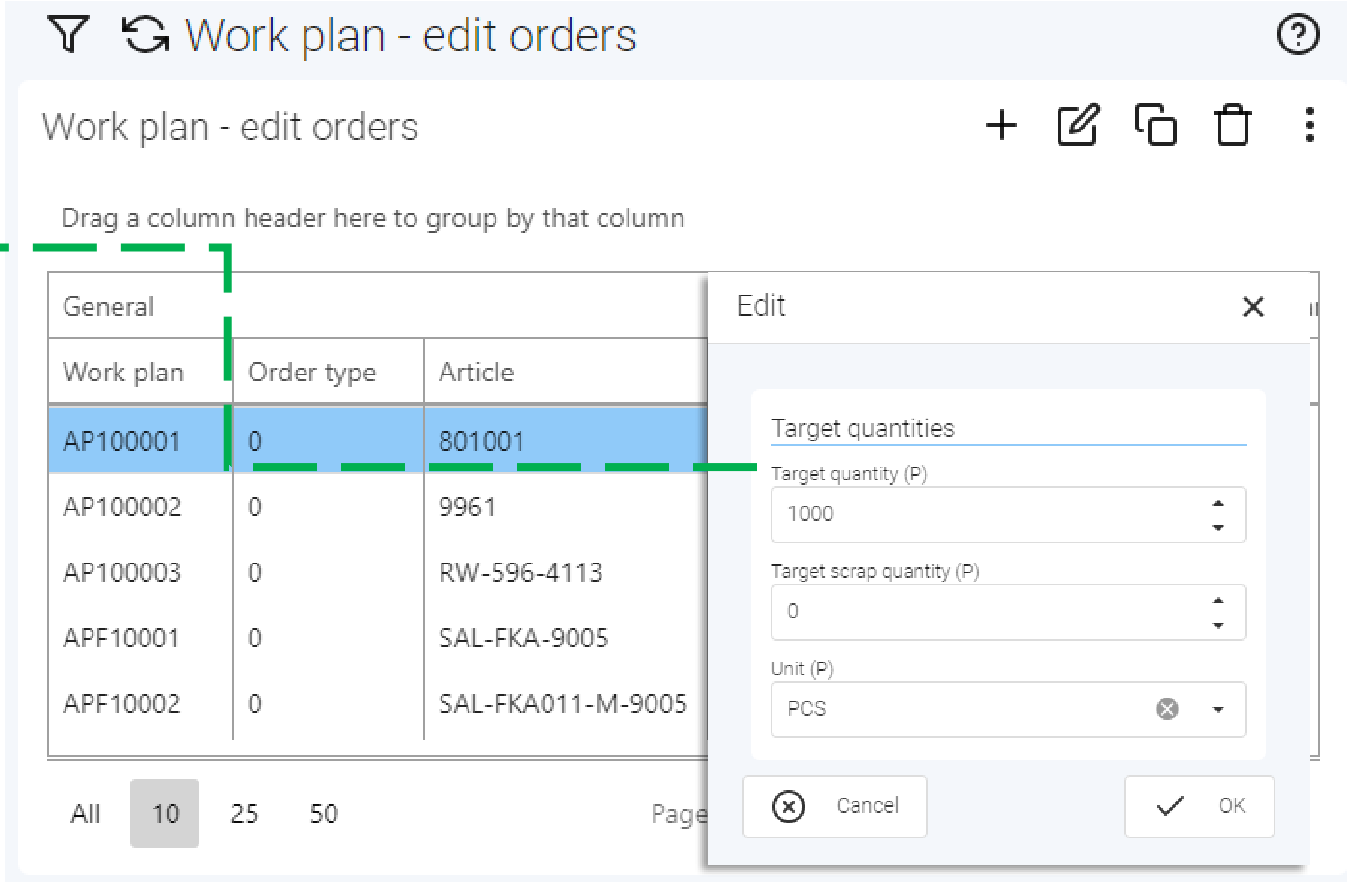
Topic 10 (continued…)
HYDRA X
Work Order Management - Work Plans & Order Data

10.4 Operation quantities

Order management → Work order management → Routing management → Work plan – Edit operations

- Base quantity (B)
  - The *Base quantity* is the unit specified in the order header.
- Primary quantity (P)
  - Primary input quantity unit for:
    - posting dialogs on the shop floor page
    - automatically recorded quantity counters
- Secondary quantity (S)
  - Other alternative quantity unit
- Tertiary quantity (T)
  - Other alternative quantity unit

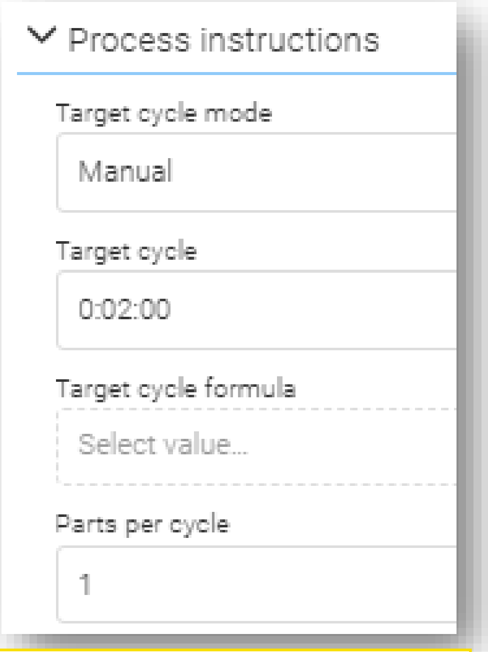


10.4 Operation quantities

Order management → Work order management → Routing management → Work plan – Edit operations

- Primary quantity (P)
  - Quantity will be based on generated orders from work plan

- In addition, it is essential / useful to indicate information such as:
  - Work plan & operation number
  - Operation Name
  - Workplace to carry out the operation ("Group" will follow your workplace of choice)
  - Processing time & Remaining run time (RRT) formula
  - Any additional timing such as queue / wait / transport time
  - Process instructions such as target cycle time and parts per cycle



Sample of 1 part per cycle and 2 minute cycle time

10.5 Time factors of operations: Target processing time

Target processing time:

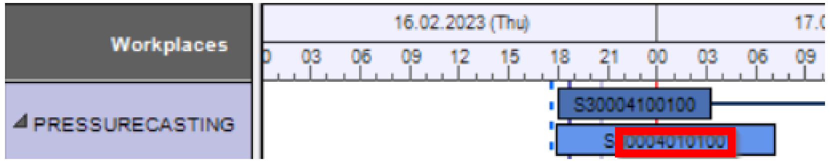
- The target processing time for the operation can either be **calculated** or stored as a **fixed** value.
  - Calculation (formula):
    - The ERP system transfers the target quantity and the target cycle as well as the parts per cylce (fixed fields in FEDRA: Mode Processing time = Formula, Formula processing time = BEA_ZY). FEDRA calculates the target processing time according to the formula **BEA_ZY**.
    - **BEA_ZY**= (Target cycle per 1000 **cycles**/1000)* target quantity/parts per cycle
  - Fix (manually):
    - The ERP system transfers the fixed value as target processing time. In this case, it can also happen that the ERP system has already calculated "Target cycle * quantity" and transfers the result to FEDRA.

Note:

The target cycle must be transferred in FEDRA (application "Edit operations") in hours per 1000 cycles. With a parts per cycle of 1, the value is equivalent to hours for every 1000 pieces.

10.5 Time Components of Operations: Length of OP Bar

Remaining run time (= *Bearbeiten* (edit)



However, the length of the bar (excl. setup and teardown time) in the FEDRA client depends only marginally on the target processing time, since the remaining run time is used for this bar. The Rem. run time is calculated with the Remaining run time formula (Field "Formula RLZ1") from the respective operation.

Normally, one of the following two formulas are used:

- **RLFZ** – Remaining run time (based on cycles):
  - (Target cycle per 1000 cycles / 1000) * ((Target quantity (P) – yield (P))/ Parts per cycle of the operation))
  - ➢ In this case, the bar becomes smaller the more yield is displayed on the terminal.
- **RLFZ_ZO** – Remaining run time based on time
  - **Target processing time** – Production time (RPA11)
  - ➢ In this case, the bar becomes shorter the longer the operation is logged on in production status.

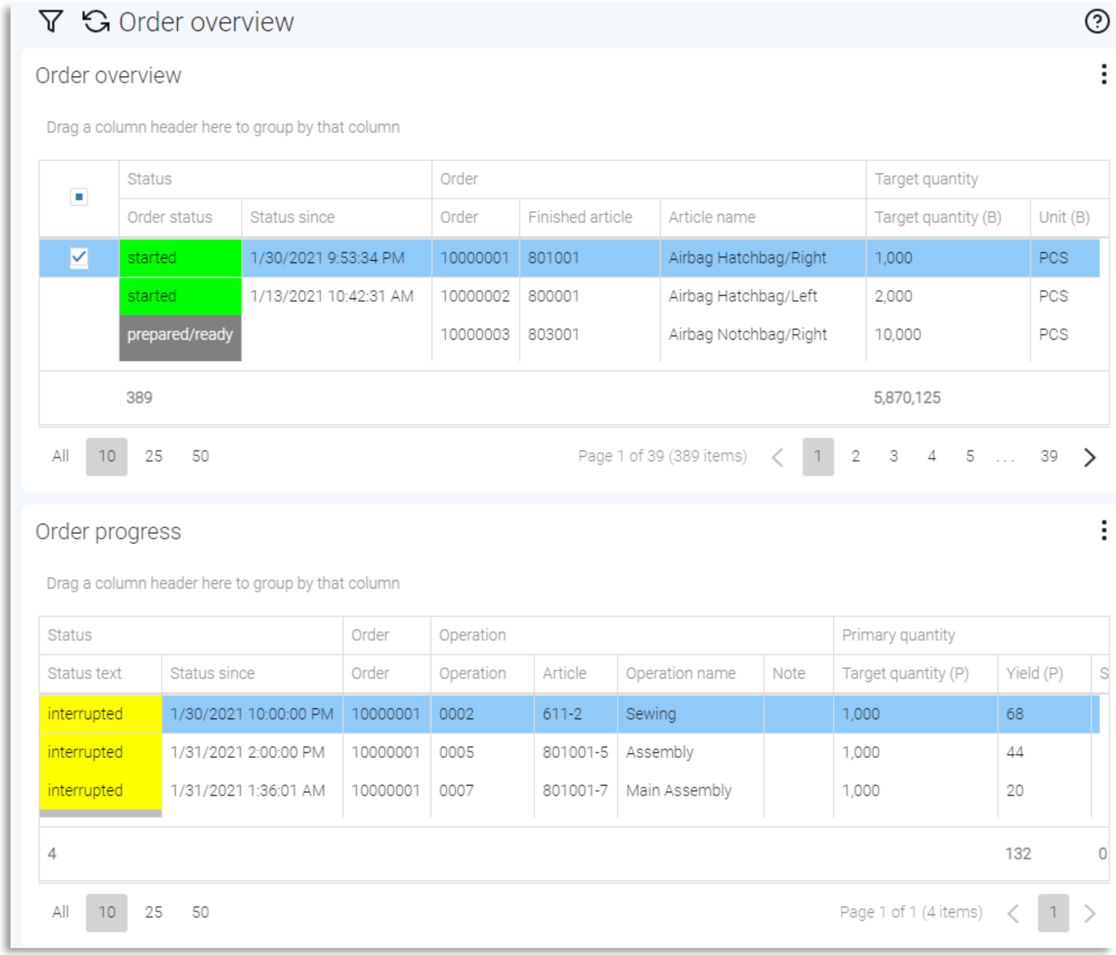
10.5 Time Components of Operations: Length of OP Bar

- Based on Cycles
  - Therefore, if the progress depends on the produced quantity, ***RLFZ*** *- Remaining run time based on cycles* must be used.
  - This is the case, for example, during punching, whenever a part is produced and a progress is recorded.
- Based on time
  - If the progress depends on the logged on duration, ***RLFZ_ZO*** *- Remaining run time based on time* must be used.
  - This is the case, for example, with an oven. Here, the quantity produced does not matter, because the drying program must run for a set period of two hours. When the operation is logged on and the status of the machine is set to *Production*, progress is recorded.
  - Consequently, only in the case of Remaining run time based on time, the target processing time of the operation is relevant for the bar length.

Note: The Remaining run time is the basis for scheduling.

10.6 Order data: Order overview

Order management → Work order execution → Order overview


Order overview

Order overview

Drag a column header here to group by that column

	Status		Order			Target quantity	
	Order status	Status since	Order	Finished article	Article name	Target quantity (B)	Unit (B)
☑	started	1/30/2021 9:53:34 PM	10000001	801001	Airbag Hatchbag/Right	1,000	PCS
	started	1/13/2021 10:42:31 AM	10000002	800001	Airbag Hatchbag/Left	2,000	PCS
	prepared/ready		10000003	803001	Airbag Notchbag/Right	10,000	PCS
	389					5,870,125	

All 10 25 50 Page 1 of 39 (389 items) 1 2 3 4 5 ... 39

Order progress

Drag a column header here to group by that column

Status		Order	Operation				Primary quantity		
Status text	Status since	Order	Operation	Article	Operation name	Note	Target quantity (P)	Yield (P)	S
interrupted	1/30/2021 10:00:00 PM	10000001	0002	611-2	Sewing		1,000	68	
interrupted	1/31/2021 2:00:00 PM	10000001	0005	801001-5	Assembly		1,000	44	
interrupted	1/31/2021 1:36:01 AM	10000001	0007	801001-7	Main Assembly		1,000	20	
4								132	0

All 10 25 50 Page 1 of 1 (4 items) 1

- Allows an overview of the production progress of an entire order and the viewing of progress details of the different operations.
- Order header data is displayed in a table format.
- For the order selected in the *Order overview*, the second table *Order progress* shows the operations assigned to the order.
- The selected operation can also be edited using “Edit Operations” to update target quantity if necessary. (The quantity for Operation will be based on the generated order quantity)

10.6 Order data: Operations

Order management → Work order execution → Operations

- Shows the current progress of all Operations (OPs)
- Useful for shift supervisors, foremen and supervisors in general for overview/monitoring purpose
- All operations are displayed in a table with its respective statuses e.g. running, prepared, interrupted etc.
- Narrow down the displayed OPs by filtering data

Operations

Operations

Drag a column header here to group by that column

Status			Order			
Status text	Status since date	Status since time	Order	Operation	Article	Operation name
interrupted	1/30/2021	10:00:00 PM	10000001	0002	611-2	Sewing
interrupted	1/31/2021	2:00:00 PM	10000001	0005	801001-5	Assembly
interrupted	1/31/2021	1:36:01 AM	10000001	0007	801001-7	Main Assembly
prepared/ready			10000001	0009	801001	Final Assembly
interrupted	3/2/2021	9:30:05 AM	10000002	0002	611-2	Sewing
interrupted	2/2/2021	10:00:00 PM	10000002	0004	800001-4	Assembly
interrupted	1/13/2021	10:50:24 AM	10000002	0006	800001-6	Main Assembly
prepared/ready			10000002	0008	800001	Final Assembly
979						

All 10 25 50 Page 1 of 98 (979 items) 1 2 3 4 5 ... 98

10.6 Order data: Operations – Filter

Order management → Work order execution → Operations

The overview contains extensive filter options e.g order number, workplaces, article types, operation statuses, etc to filter.

Enter your filter criteria before requesting data.

Filter

- Workplace
- Group
- Planned for
- Order
- Operation
- Article
- Operation status
- Last logon/logoff
- Earliest start and end
- Latest end and start
- …

10.6 Order data: Persons logged on

Order management → Work order execution → Persons logged on

The overview *Persons logged on* shows a list of persons currently logged on to operations and workplace.

(Useful to see how the workers are utilised so as to reschedule or replan the manpower, as needed)

Persons logged on

Drag a column header here to group by that column

Person		Login		Workplace	Order	
Person	Name	Started on	Duration	Workplace	Order	Ope
999998	Meier, Hans	6/10/2021 6:21:03 PM	14:12:23	60610		
59881	Brown, Albert	6/10/2021 6:23:44 PM	14:09:42	60612		
999999	Schulz, Werner	6/10/2021 6:24:45 PM	14:08:41	60614		

All 10 25 50

Page 1 of 1 (3 items) 1

10.7 Order sequencing: Table

Order management → Work order management → Production support → Order sequencing

In the MES system, there is the option to reorder the sequence of operations from planned orders such as production orders to be executed on various workplaces.

- Useful tool for planners, shift foremen and supervisors
- Plan the sequence of operations on machines
- Called “Order sequencing” In Hydra MES, an application of the production support to allow planning the operation sequence via drag and drop. However, it does not account for any possible constraints.

The order sequencing application includes the following functions:

- Display operations stored in the group backlog
- Display operations planned for workplaces (under workplace backlog)
- Allow a processing sequence ("sequencing") to be defined – this specifies how the operations to be produced are displayed in the sequencing list in a specific order.
- Plan an operation for a workplace/machine

10.7 Order sequencing: Table

The Order sequencing application includes three tables:

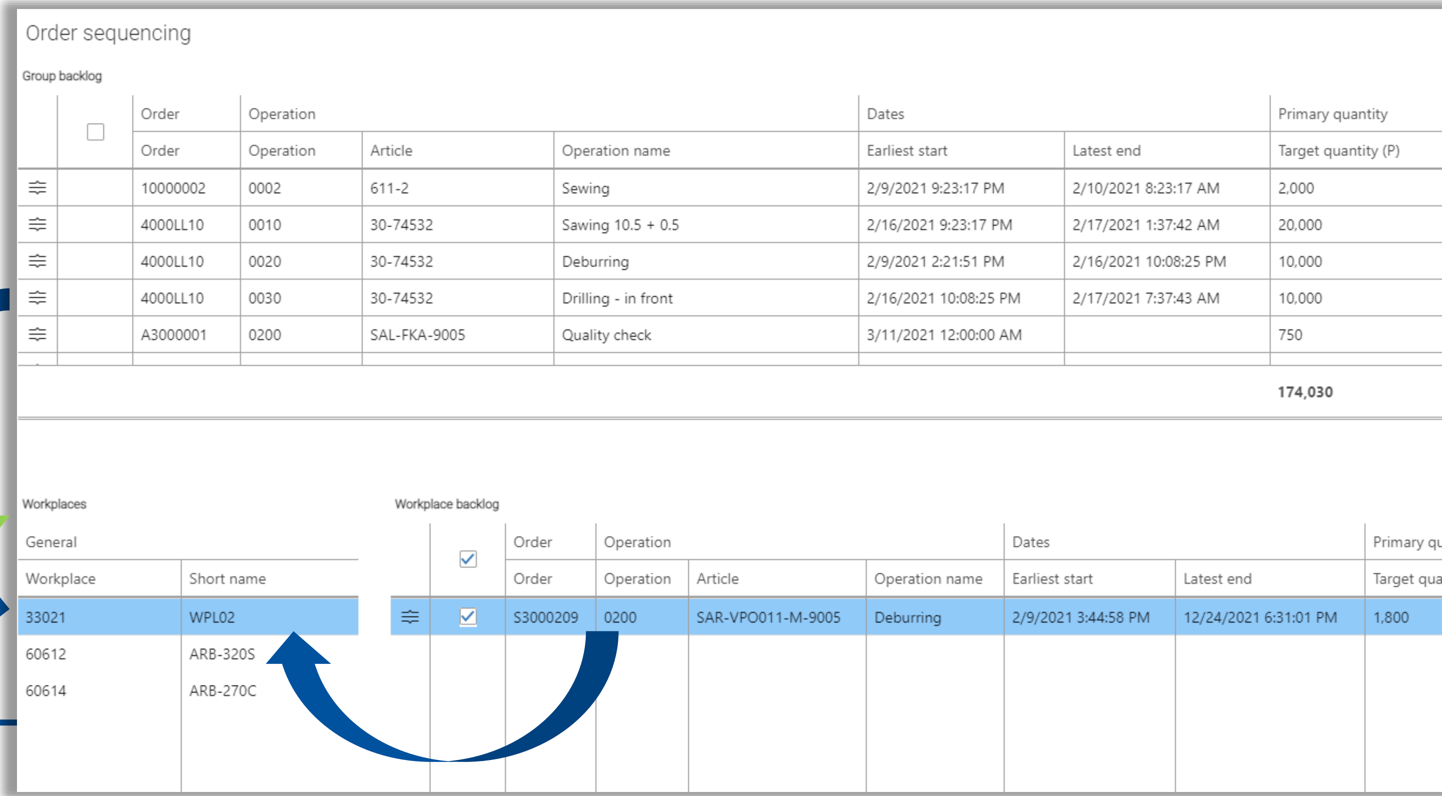
1. Group backlog
2. Workplaces
3. Workplace backlog

Order management → Work order management → Production support → Order sequencing

In the "order sequencing" app, user will be able to view and sequence the orders according to group backlogs and workplace backlogs.

Do you recall?

- Flexible configuration to configure your workplace for order sequencing list in terms of group or workplace
- Additional configuration (workplace sequencing list count) to see only a specific number of operations on the shopfloor device (sequencing list) e.g. 50 operations


Order sequencing

Group backlog

	Order	Operation			Dates		Primary quantity
☐	Order	Operation	Article	Operation name	Earliest start	Latest end	Target quantity (P)
	10000002	0002	611-2	Sewing	2/9/2021 9:23:17 PM	2/10/2021 8:23:17 AM	2,000
	4000LL10	0010	30-74532	Sawing 10.5 + 0.5	2/16/2021 9:23:17 PM	2/17/2021 1:37:42 AM	20,000
	4000LL10	0020	30-74532	Deburring	2/9/2021 2:21:51 PM	2/16/2021 10:08:25 PM	10,000
	4000LL10	0030	30-74532	Drilling - in front	2/16/2021 10:08:25 PM	2/17/2021 7:37:43 AM	10,000
	A3000001	0200	SAL-FKA-9005	Quality check	3/11/2021 12:00:00 AM		750
							174,030

Workplaces

General	
Workplace	Short name
33021	WPL02
60612	ARB-320S
60614	ARB-270C

Workplace backlog

	Order	Operation			Dates		Primary qu
☑	Order	Operation	Article	Operation name	Earliest start	Latest end	Target qua
☑	S3000209	0200	SAR-VPO011-M-9005	Deburring	2/9/2021 3:44:58 PM	12/24/2021 6:31:01 PM	1,800

Workplaces
Selecting a workplace allows the viewing of operations planned for this workplace (under "Workplace backlog")

** Workplaces meeting the following criteria are shown if :

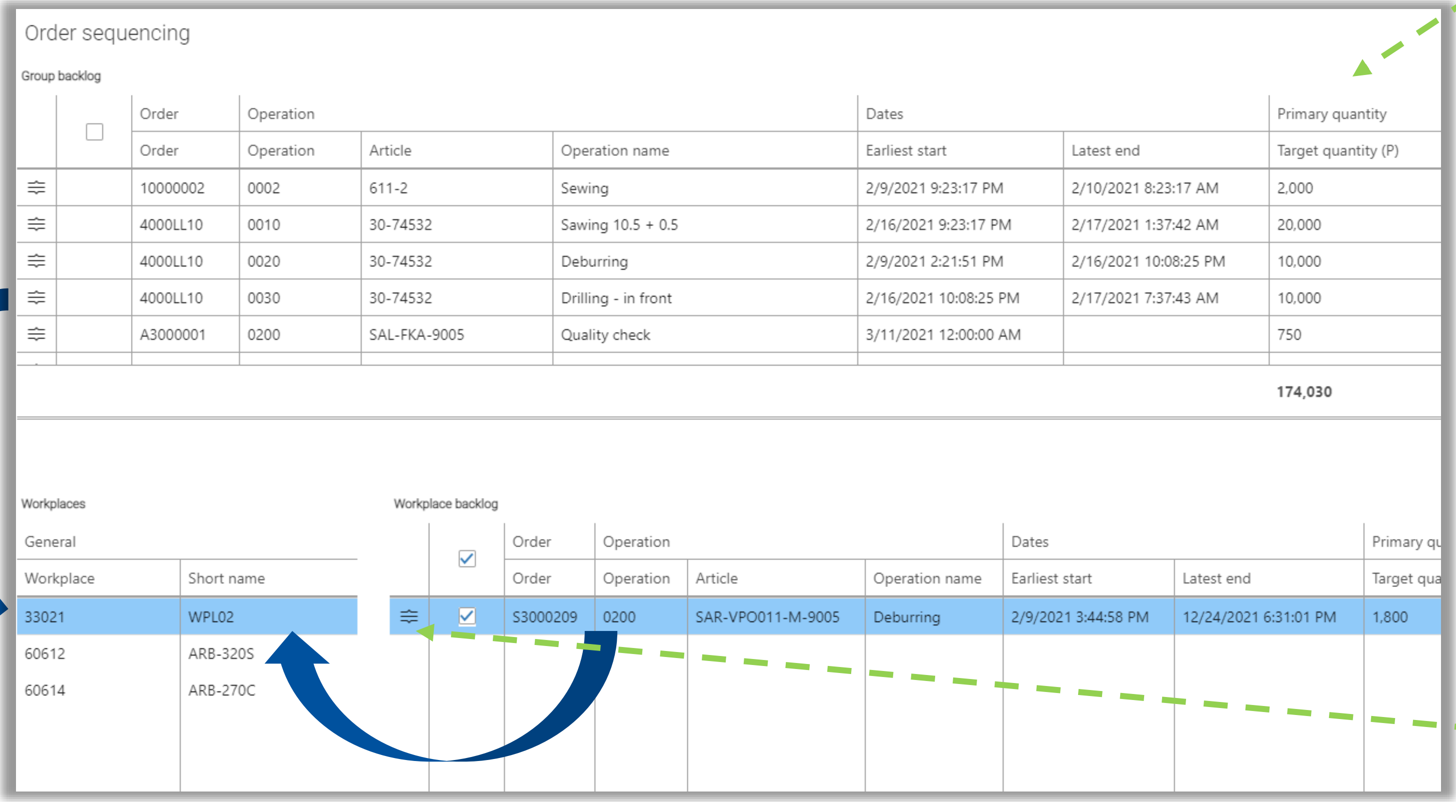
- Planning function is set to Planning in order sequencing.
- The planner is authorized for the responsibility area of the workplace(s)

Official (Open)

10.7 Order sequencing: Table

Order management → Work order management → Production support → Order sequencing

Operation(s) in the group backlog that are to be planned for a workplace


Order sequencing

Group backlog

		Order	Operation			Dates		Primary quantity
		Order	Operation	Article	Operation name	Earliest start	Latest end	Target quantity (P)
		10000002	0002	611-2	Sewing	2/9/2021 9:23:17 PM	2/10/2021 8:23:17 AM	2,000
		4000LL10	0010	30-74532	Sawing 10.5 + 0.5	2/16/2021 9:23:17 PM	2/17/2021 1:37:42 AM	20,000
		4000LL10	0020	30-74532	Deburring	2/9/2021 2:21:51 PM	2/16/2021 10:08:25 PM	10,000
		4000LL10	0030	30-74532	Drilling - in front	2/16/2021 10:08:25 PM	2/17/2021 7:37:43 AM	10,000
		A3000001	0200	SAL-FKA-9005	Quality check	3/11/2021 12:00:00 AM		750
								174,030

Workplaces

General	
Workplace	Short name
33021	WPL02
60612	ARB-320S
60614	ARB-270C

Workplace backlog

		Order	Operation			Dates		Primary qu
		Order	Operation	Article	Operation name	Earliest start	Latest end	Target qua
		S3000209	0200	SAR-VPO011-M-9005	Deburring	2/9/2021 3:44:58 PM	12/24/2021 6:31:01 PM	1,800

Workplace backlog

- Sequence of operations planned for the workplace can be changed by dragging and dropping the operation in the order that it shall be scheduled, should there be multiple operations.
  (Order has to be generated first!)
- Operations will be displayed in the same sorted sequence in the sequencing list under Shopfloor OM/RM page.

10.7 Order sequencing: Workplace configuration

MIP Administration → Administration → Flexible configuration

Flexible configuration — planning function	
Object type	Workplace
Parameter name	WorkplacePlanningFunction
Search rule (to select)	Choose one of the following based on requirement and indicate the relevant inputs as identifier: ***workplace.*** **<*id*> - *workplace number*** **<*Group*>** **<*Category*>** **<*Cost center*>** **<*Company*>**
Value	E.g. Indicate workplace number if workplace.id is chosen
X value	N | F | S
Data type	char *(character)*

Priority

N = no planning
F = planning in FEDRA
S = planning in tabular order sequencing (in Hydra MES)

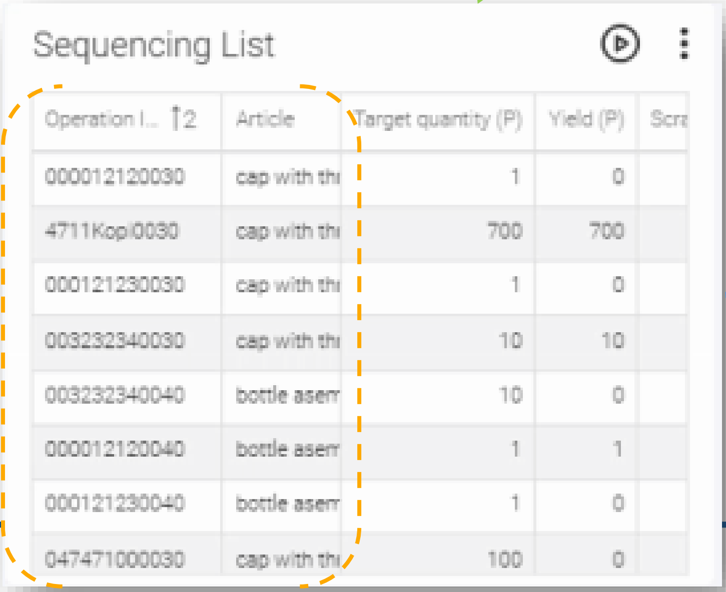
Machine is being evaluated

- ❑ input method similar to previous flexible configuration (s)
- ❑ flexible configuration to allow viewing of the workplace in ‘Order Sequencing’ application
- ❑ planning options available:
  - ❑ Hydra MES system
  - ❑ FEDRA (type of Advanced Planning and Scheduling System)
  - ❑ No planning

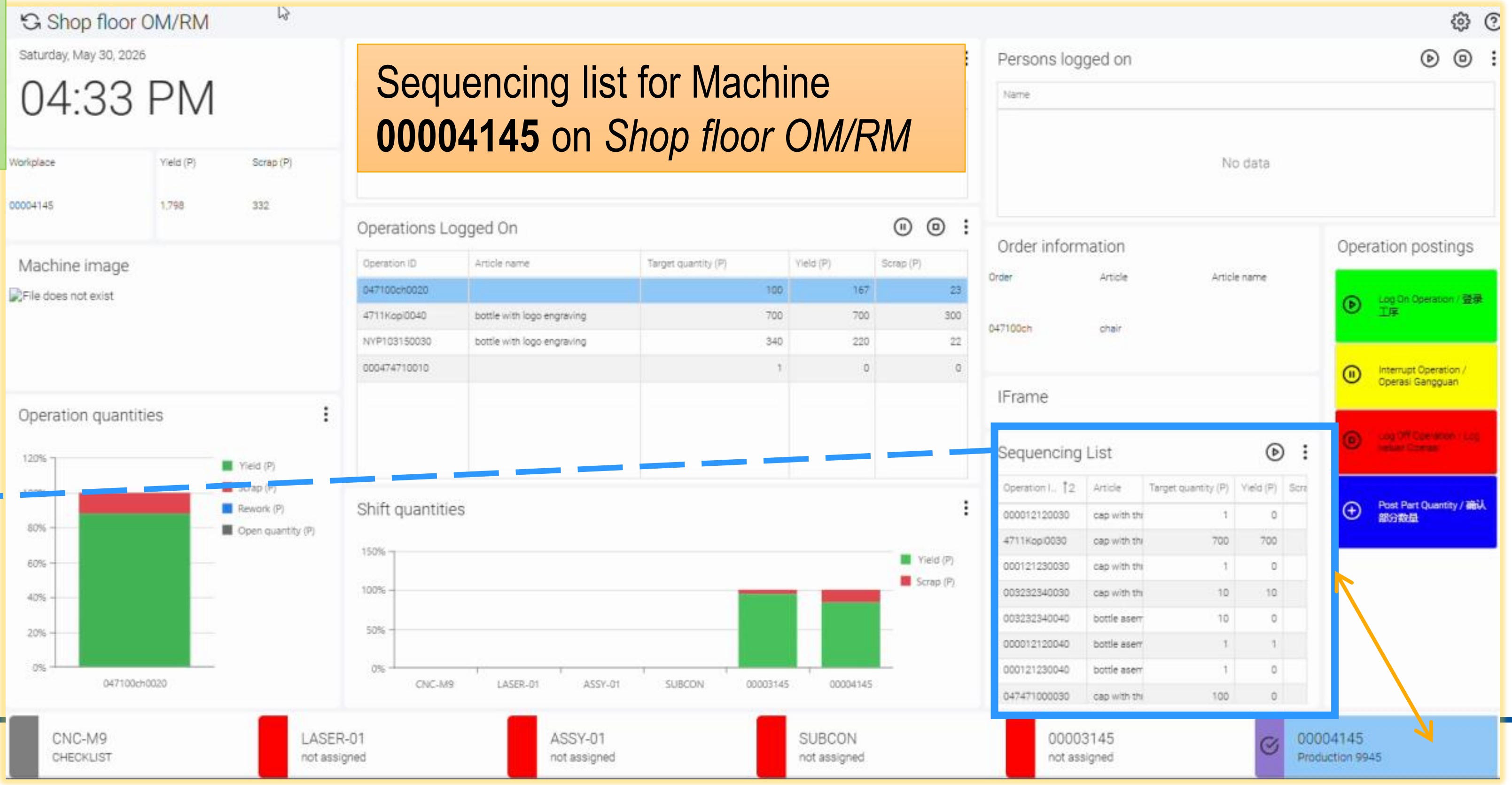
10.7 Order sequencing & Shop floor OM/RM (Sequencing list)

Workplace backlog for Machine **00004145** on *Order Sequencing* Application

Order of workplace backlog for Machine **00004145** will be reflected on *Order Sequencing* Application to allow for logging on of operations



Sequencing list for Machine **00004145** on *Shop floor OM/RM*



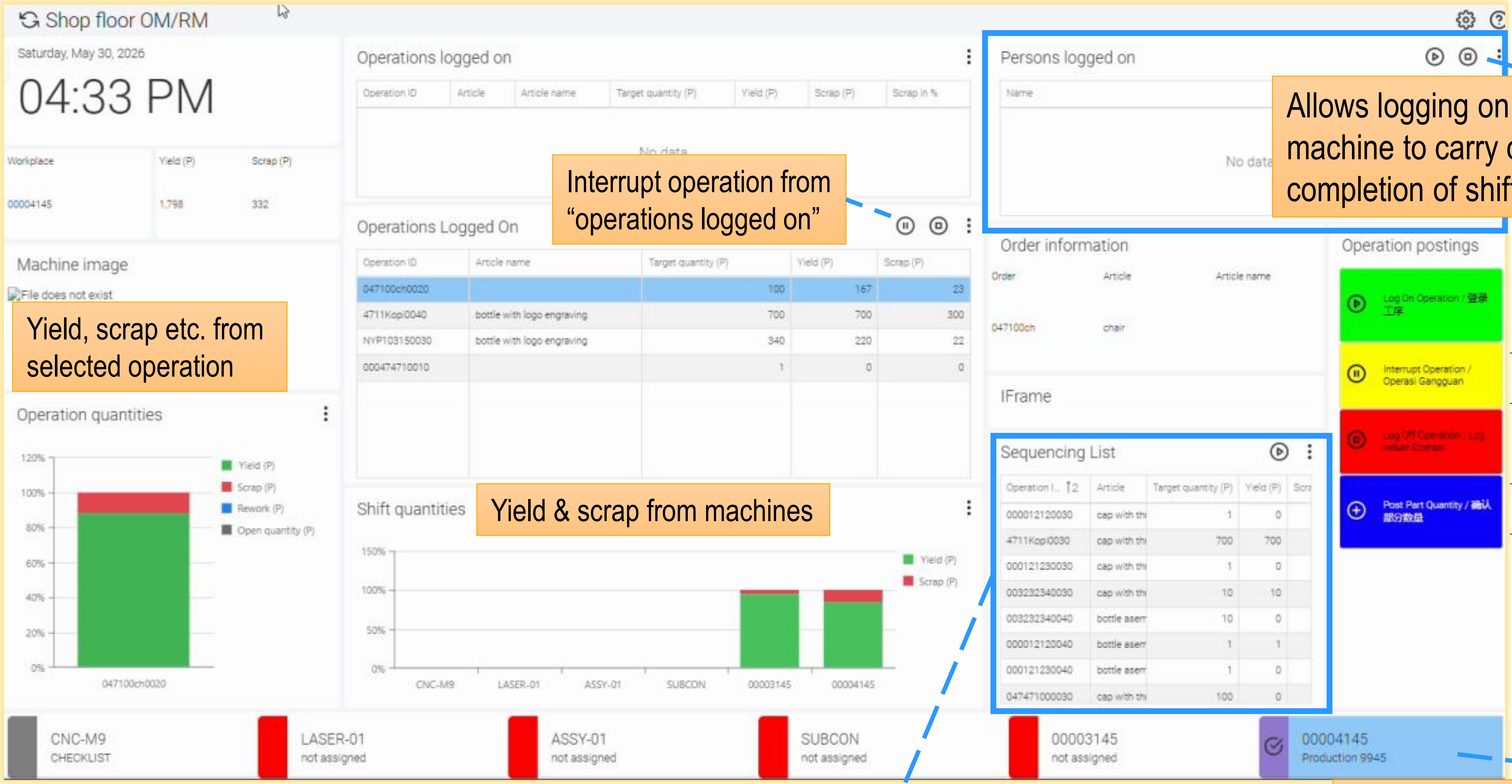
10.8 Shop floor OM/RM

The Shop floor OM/RM page can be used to process the generated orders from the work plan:

- logging on / off of persons
- logging on of operation
- posting of part quantity for operation
- interrupt operation (e.g. when “logged on operations” is completed)
- viewing of operations on sequencing list
- viewing of yield / scrap from operations or workplace
- viewing the logged on operation(s)

Illustration on next page …

10.8 Shop floor OM/RM



Allows logging on / off of persons to the machine to carry out the operation / completion of shift (key in card ID e.g. 8845)

Interrupt operation from "operations logged on"

Yield, scrap etc. from selected operation

Yield & scrap from machines

Log on operation from "sequencing list"

Interrupt operation from "operations logged on"

Post part quantities under "operations logged on"
- *Good parts under yield*
- *Rejects under scrap with damage reasons*

List of planned operations in arranged sequence from generated orders

Select Machine for viewing or control e.g. logging on of operation / persons

10.9 Overdelivery / Underdelivery Example

Underdelivery (Deviation value in %)

- If the target quantity of the operation is 100%, a value greater than 100% is not permitted with underdelivery.
- Example: For a target operation quantity of 120 pieces with underdelivery of 84%, the actual quantity must not fall below 101 pieces.

120 pieces ∗ 84% = 100.8 pieces *[minimum quantity]*
→ *shall not be less than 101 pieces in this case,* after considering a whole number but not less than the calculated

Overdelivery (Deviation value in %)

- If the target quantity of the operation is 100%, a value less than 100% is not permitted with overdelivery.
- Example: For a target operation quantity of 120 pieces with overdelivery of 168%, the actual quantity must not exceed 201 items.

120 pieces ∗ 168% = 201.6 pieces *[maximum quantity]*
→ *shall not be more than 201 pieces in this case, after considering a whole number but not more than the calculated*

Overdelivery reaction / underdelivery reaction

A "warning" or "error" can be output as reaction in the MES system for exceeding the limits specified

- **Sufficient**: No reaction in the event of an overdelivery or underdelivery
- **Warning**: Warning if tolerance quantity is exceeded or not reached

Exercise: Generate work plan and order

Exercise requires “99XX” account

Replace XX with last 2 digits of your login ID: 9934

e.g. BOTTLE34

Generate an order to produce a bottle

Order management → Work order management → Routing management → Work plan – Edit orders

1. Familiarize yourself with the work plan for bottle & create one work plan: "BOTTLE**XX**“

Order management → Work order management → Routing management → Work plan – Edit operations

2. Familiarize yourself with the operations of the bottle work plan & add them to the work plan "BOTTLE**XX**“ *(1 complete bottle comprises of 1 bottle & 1 cap)*

3. Add in tolerance info for one of the individual operations of the work plan using “Work plan – Edit long texts of OP”

- Operation: Injection Molding
- Length tolerance: 17.0 +/-0.6 mm

Work plan: 00004711
Finished article: bottle

- OP 10: injection molding
- OP 20: punching
- OP 30: thread milling
- OP 40: assembly & engraving

Exercise: Generate work plan and order

Order management → Work order management →Routing management → Work plan - Edit orders

4. Use the function *Generate order* to generate an order.
   - Enter the *Basic date start*.
   - The order number is “TRAIN0**XX**”
   - Enter the *quantity e.g. 300*
   - *Enter the customer's name*

Order management → Work order management → Routing management → Work plan – Edit orders → Edit Operations

5. Try creating “overdelivery” of 120% for operation 0010 of your generated order with a reaction of “warning”

6. Open the function “*Order overview*” and check your new order.
   - The operations linked to the selected order will be displayed
   - (User may edit the quantity for specific operations as needed)

Exercise: Order sequencing

Exercise requires “99XX” account

Use the order sequencing to change the OP sequence on the device.

Order management → Routing management → Work plan - Edit orders

1. Generate new orders "BOTTL**XX1**“, "BOTTL**XX2**“… from the work plan "BOTTLE**XX**“ or sample work plan “**00004711**”, while indicating details such as quantity e.g. *300, 500*

   (Sample machines used in work plan “00004711”: “00003145” & “00004145”)

MIP Administration → Administration → Flexible configuration

2. Activate the option *Planning in order sequencing* for all your machines.

Production control → Production support → Order sequencing

3. Look for the operations from your generated order under the relevant workplace(s) e.g. 000041**XX**. View & schedule the created OPs for your workplace.
4. Reorder (drag up/down) the sequence of your OPs in the order sequencing dialog and check the changes to the sequencing list on the shop floor page (* Do remember to refresh there; refresh icon under “3 dots” menu).

Optional Exercise: Order sequencing (cont'd) – Shop floor OM / RM

After completing the changing of OP sequence on the device, further steps can be processed on the shop floor page (Shop floor OM / RM**)**

1. Log on staff after selecting the required machine. (Valid Card ID is required e.g. 8845)
2. Log on operation from the sequencing list via "Log on Operation" (Check "Operations Logged On" for updates)
3. Now any yield / scrap can be posted via "Post Part Quantity" to the selected operation for a machine. (A reason can be issued for scrap)
4. After completion, the staff / operation can then be logged off

For this exercise,
- Either have a look using account "sfs45" to view the Shop floor OM / RM with more widgets setup / buttons control
- Or add machines to own "99XX" account to view the Shop floor OM / RM

Summary: Work Order Management (WOM)

Client functions

- The *Order overview* shows order and operation information on a screen that is separated into 2 windows namely,:
  - 'Order Overview' to see the statuses of all the ORDERS
  - 'Order Progress' to see the statuses of all the OPERATIONS within the orders

- The *Order information* is the central screen and shows:
  - master data of an order and its operations
  - current status of orders and operations and further details.

- The overview *Persons logged on* shows a list of staff currently logged on.

- Possible to add fields to pivot tables in HYDRA - The charts vary according to the selected fields.

Summary / Additional Information

- The option to allow for generating different orders in MES system caters for different purposes such as production order for customers, overhead orders to collect time instead of product quantity etc.

- Application “Work Plan - Edit Production Resources & Tools” allows the displaying and editing of the resources, which are required to produce the product in the specific operation). For example, production resources and tools may be tools, documents, NC programs, etc.

- Application “Work Plan - Edit Operation Long Texts” allows for additional texts to be collected for the operation. The terminal shows the long texts relating to operations.

- Application “Work Plan - Edit Components displays and edits the material components, which are required to produce the article in the current manufacturing level (current operation). Normally, these components are transferred to this system using an interface from the higher-level ERP system, as these components are already defined in the ERP work plan.